Customer Shopping Behavior Analysis

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1. Project Overview

This project analyzes customer shopping behavior using transactional retail data consisting of 3,900 customer records. The objective was to uncover patterns in revenue generation, customer segmentation, subscription behavior, and product performance to support data-driven business decisions.

The analysis was conducted using Python for data cleaning and feature engineering, SQL for business analysis, and Power BI for dashboard visualization.

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2. Dataset Summary

- Total Rows: 3,900
- Total Columns: 18 (after cleaning: 17)
- Data Type: Transaction-level retail dataset

Key Features

Customer Information:

- Age
- Gender
- Location
- Subscription Status

Purchase Details:

- Item Purchased
- Category
- Purchase Amount (USD)
- Season
- Size
- Color

Behavioral Metrics:

- Discount Applied
- Previous Purchases
- Frequency of Purchases
- Review Rating

- Shipping Type

Data Quality

- 37 missing values identified in the Review Rating column
- No duplicate records detected
- Column names standardized to snake_case format

(Insert dataset preview screenshot here)

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3. Data Cleaning & Feature Engineering (Python)

Data preparation was performed using Python (pandas) to ensure consistency and analytical readiness.

Key Steps Performed:

- Loaded dataset and explored structure using df.info() and df.describe()
- Identified missing values and handled 37 null values in Review Rating
- Imputed missing ratings using category-wise median values
- Standardized column names to lowercase with underscores
- Created age_group using quartile-based segmentation (pd.qcut) for balanced demographic analysis
- Converted purchase frequency categories into numerical values (purchase_frequency_days)
- Verified redundancy between promo_code_used and discount_applied
- Dropped redundant column (promo_code_used)
- Exported cleaned dataset for SQL and Power BI analysis

```
df.describe(include = 'all')
```

Show hidden output

```
df.isnull().sum()
```

Show hidden output

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x : x.fillna(x.median()))
```

```
df.isnull().sum()
```

Show hidden output

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ','_')
```

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4. Business Analysis Using SQL

Structured SQL queries were written to answer strategic business questions.

4.1 Revenue & Customer Insights

- Revenue distribution across gender and age groups
- Comparison of average spend and total revenue between subscribers and non-subscribers
- Pareto analysis to evaluate revenue concentration

Key Finding:
The Pareto analysis revealed that the top 20% of customers contribute approximately **32.54%** of total revenue, indicating a relatively balanced revenue distribution rather than a concentrated 80/20 pattern.

gender	age_group	revenue	revenue_percentage
Female	Young Adult	20142	8.64
Female	Middle-aged	19806	8.50
Female	Adult	17759	7.62
Female	Senior	17484	7.50
Male	Young Adult	42001	18.02
Male	Middle-aged	39391	16.90
Male	Senior	38279	16.42
Male	Adult	38219	16.40

total_customers	subscription_status	avg_spend	total_revenue	revenue_percentage
2847	No	59.87	170436	73.12
1053	Yes	59.49	62645	26.88

Result Grid | Filter Rows:

top_20_percent_revenue_share
32.54

4.2 Product Performance Analysis

- Top 5 products with highest average review ratings
- Top 3 most purchased products within each category
- Products with highest percentage of discounted purchases

Key Findings:

- Gloves, Sandals, and Boots have the highest average customer ratings, indicating strong satisfaction levels.
- Blouse (Clothing), Jewelry (Accessories), Sandals (Footwear), and Jacket (Outerwear) rank as the most purchased products within their categories.
- Products such as Hat and Sneakers show high discount dependency (~50%), suggesting promotions significantly influence their sales.

item_purchased	average_review_rating	total_reviews
Gloves	3.86	140
Sandals	3.84	160
Boots	3.82	144
Hat	3.8	154
Skirt	3.78	158

item_rank	category	total_orders	item_purchased
1	Accessories	171	Jewelry
2	Accessories	161	Sunglasses
3	Accessories	161	Belt
1	Clothing	171	Blouse
2	Clothing	171	Pants
3	Clothing	169	Shirt
1	Footwear	160	Sandals
2	Footwear	150	Shoes
3	Footwear	145	Sneakers
1	Outerwear	163	Jacket
2	Outerwear	161	Coat

item_purchased	total_purchases	discount_rate
Hat	154	50.00
Sneakers	145	49.66
Coat	161	49.07
Sweater	164	48.17
Pants	171	47.37

4.3 Behavioral & Operational Analysis

- Comparison of average purchase value between discounted and non-discounted purchases
- Shipping type analysis (Standard vs Express)
- Relationship between repeat buyers and subscription status

Key Finding:
Discount usage and shipping type influence purchasing patterns, offering opportunities for pricing and operational optimization.

discount_applied	total_orders	avg_order_value
Yes	1677	59.28
No	2223	60.13

shipping_type	total_purchases	avg_purchase_amount
Express	646	60.48
Standard	654	58.46

subscription_status	repeated_buyers
No	2518
Yes	958

4.4 Customer Segmentation

Customers were segmented based on previous purchase behavior:

- New (1 purchase)
- Returning (2–10 purchases)
- Loyal (>10 purchases)

This segmentation supports targeted marketing and retention strategies.

customer_segment	total_customers
Loyal	3116
Returning	701
New	83

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5. Power BI Dashboard

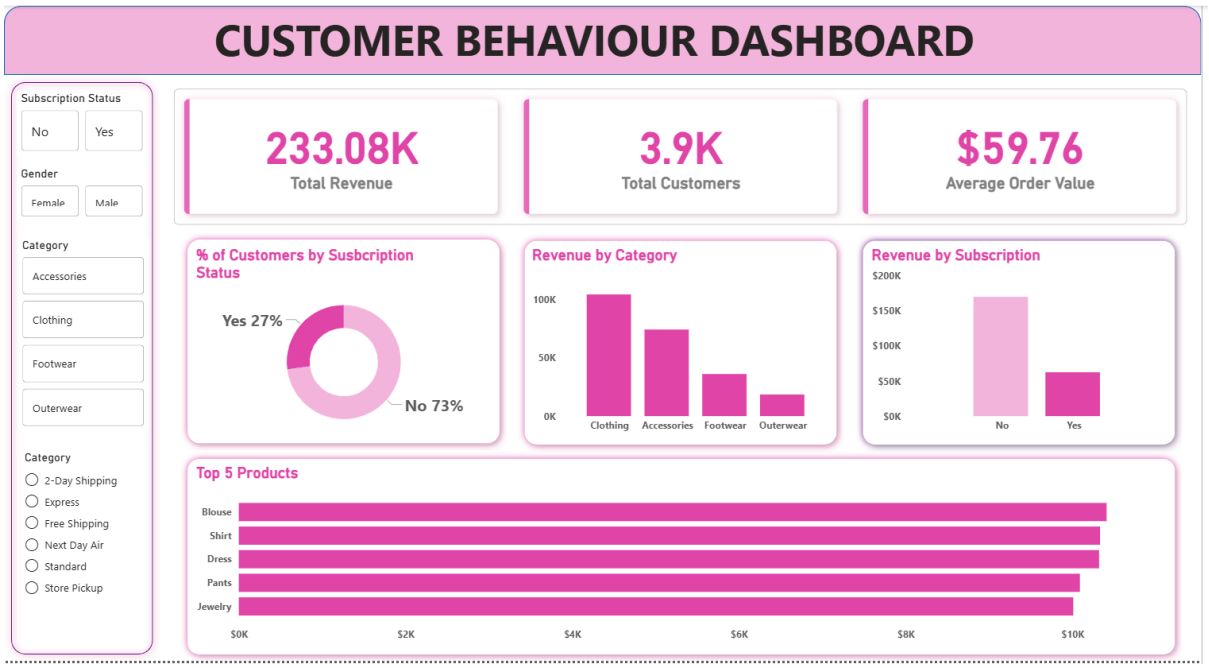
An interactive dashboard was developed to visually communicate insights.

Key KPIs:

- Total Revenue: $233.08K
- Total Customers: 3,900
- Average Order Value: $59.76

Dashboard Highlights:

- 73% of customers are non-subscribers
- Clothing is the top revenue-generating category
- Non-subscribers contribute higher total revenue due to larger customer base
- Top 5 products drive a significant portion of category-level revenue



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6. Business Recommendations

Based on the analysis, the following strategic recommendations are proposed:

1. Enhance Subscription Strategy

Strengthen subscription benefits (exclusive discounts, loyalty rewards) to increase adoption and long-term retention.

2. Prioritize High-Performing Categories

Focus marketing efforts on high-revenue categories such as Clothing while identifying growth opportunities in underperforming categories.

3. Promote Top Revenue-Driving Products

Highlight products like Blouse, Shirt, and Dress in campaigns and bundled promotions to sustain consistent sales performance.

4. Optimize Discount Strategy

Evaluate discount-heavy products to ensure promotions drive incremental revenue without negatively impacting profit margins.

5. Maintain Broad Customer Engagement

Since revenue distribution is relatively balanced (Top 20% contribute ~32.54%), growth strategies should focus on engaging the broader customer base rather than relying solely on high-value customers.

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7. Conclusion

This project demonstrates an end-to-end analytics workflow, integrating Python for data preparation, SQL for structured business analysis, and Power BI for visual storytelling.

The findings provide actionable insights into customer behavior, revenue drivers, subscription impact, and product performance, enabling data-driven strategic decision-making.